

Warming Up Fairly: Language Models for Cold-Start Recommendation and Provider Equity

University of Michigan - Master of Applied Data Science: Capstone Project

Team Cold Start (Group 11), Spring/Summer 2026

<https://github.com/jessylan/llm-cold-start-reccsys>

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Introduction and Project Statement

Cold items and cold users (without interaction history) are a structural problem for recommender systems. Pure **collaborative filtering (CF)** models depend on interaction history and cannot rank cold items. Adding content vectors to make use of product metadata is an established practice in cold-start literature [1-7], but interaction history remains critical for quality recommendations. This is the **cold-start problem**, which we set out to measure and improve, with a focus on cold items.

We evaluate recovery rather than a single snapshot, adding interactions one at a time to trace a **warm-up curve** showing how quickly an item becomes findable. We measure this curve from two directions: **Mode A** ranks items for a user, which is the familiar recommendation task, and **Mode B** recommends users for an item, informing publisher tasks such as targeted marketing.

We create three baseline models per mode, each adding capability above the last. Naïve **Popularity** and **Activity** baselines establish the floors. A pure CF model (Alternating Least Squares, **ALS**) shows what interaction data alone achieves. Our true baseline is the content-based hybrid collaborative filtering algorithm (**CBHCF**), which adds TF-IDF content embeddings to the ALS model, allowing cold items to be ranked based on content before any interactions happen.

We then test two interventions. **Intervention A** replaces the CBHCF model's TF-IDF content embeddings with dense semantic text embeddings. **Intervention B** adds LLM-generated synthetic interactions. Each intervention changes one thing and holds the rest fixed.

Accuracy is not the only thing worth measuring. An improved recommender could lift the warm-up curve while still concentrating exposure on the same handful of authors. We therefore also track **provider-side equity**, which captures whether smaller providers and newer items get a fair shot at exposure. We compute it only for Mode A.

Appendix A describes the system in detail.

Context and Related Works

Recent work argues the transition between complete cold start and mature recommendation deserves evaluation as its own warm-up phase. AlRossais [8] notes that recommender-system research typically evaluates either completely cold or fully established users and items, which motivates our use of warm-up curves rather than relying only on aggregate scores.

The growing use of large language models (LLMs) introduces additional strategies for cold recommendation. Zhang et al. [9] organize cold-start approaches according to the external information used, and distinguish methods that use LLMs to enrich representations from methods that use LLMs to simulate behavior. Our two interventions follow this distinction: Intervention A uses semantic item representations, while Intervention B generates synthetic interactions as a form of data augmentation.

Data and Methodology

Amazon Review Dataset

The Amazon Reviews 2023 Books dataset contains user-item interactions and metadata needed for both collaborative and content-based recommendation [7, 10]. After filtering, we keep roughly 2.67 million interactions and rank over a candidate pool of 246,687 items. Appendix A describes our data and filtering in more detail.

Cold Start Methodology

We examine the cold-start problem by simulating item novelty and incrementally revealing 20 interactions per item. User preferences are fit once on the warm training set, frozen, and reused identically for every method. Only the item's representation changes as interactions accumulate. At each reveal level, every cold item is folded in and evaluated against the same fixed reserved test set, resulting in the warm-up curve. A change identified in retrieval quality is therefore attributable to the item gaining signal.

Pipeline

The pipeline has baselines and two interventions. The baseline models form a hierarchy of increasing information availability. Popularity and Activity have no user- or item-specific signal. ALS draws on interaction history alone. Our true baseline is CBHCF, which adds content signal. Each intervention branches off CBHCF.

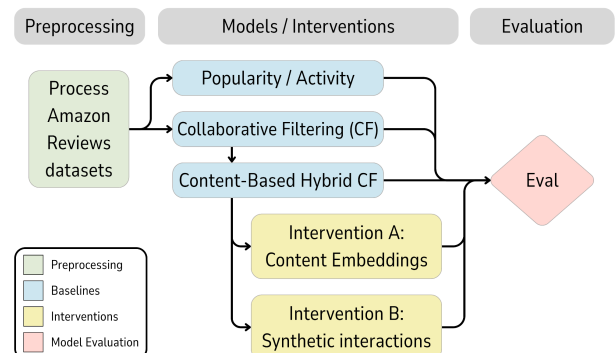


Figure 1. Cold Start Intervention Pipeline

Global Item Popularity: Popularity recommends items based on their total interaction count in the training data. It functions as a lower bound, establishing whether a more sophisticated model's added complexity earns its keep.

Global User Activity: Activity is Popularity analog for Mode B and recommends users based on total interaction count. It functions as the lower bound for Mode B.

Alternating Least Squares (ALS): ALS is the pure CF baseline. It learns latent user and item representations purely from interaction history. As it has no content fallback, items without interactions cannot be ranked.

CBHCF (Content-Based Hybrid CF): CBHCF adds TF-IDF representations of product metadata (“content”) to the user-item interaction patterns leveraged by the pure CF baseline. This enables the model to rank items even when its interaction history is limited or absent.

Intervention A (Content Embeddings): Intervention A modifies CBHCF’s content representation by replacing standard TF-IDF term-frequency vectors with LM-derived text embeddings. It attempts to capture the semantic meaning of the content information.

Intervention B (Synthetic Interactions): Intervention B uses an LLM to generate synthetic interactions for cold items. This happens in two stages. First, we identify users whose content profiles are most similar to the cold item. Then, we ask the LLM to judge whether the user would interact with the item. We order its responses and add the top seven interactions per cold item to the fold-in at each reveal level.

Evaluation and Metrics

The system is evaluated from two directions, ranking along both axes of the same user-by-item matrix. Mode A ranks items for a user and asks whether a cold item reaches that user’s top-K recommendations. Mode B ranks users for an item and asks whether the item’s ranking surfaces the users who actually wanted it. The two modes are computed by separate code paths with separate ceilings and separate list lengths ($K=100$ for Mode A, $K=10$ for Mode B). We distinguish this list length, capital K, from the 20 reveal levels written as lowercase k, which forms the x-axis of every curve we report. Both symbols appear together throughout our results, so the distinction matters.

We compute $\text{NDCG}@K$, $\text{Recall}@K$, $\text{Precision}@K$, and $\text{HitRate}@K$ at each reveal level to trace the warm-up curve, an approach grounded in AlRossais [8]. Every curve is paired with a ceiling reference: the same cold item folded in with its complete pre-test interaction history and evaluated against the same held-out test set of 5 interactions, the within-item protocol of Meta-Embedding [2] and MWUF [6]. This gives each metric an interpretable upper bound.

We evaluate provider-side equity using the same reveal-level structure, so exposure and accuracy plot on the same x-axis. Our primary equity measures are the *Gini coefficient of exposure* [11, 12], where 0 indicates perfectly equal exposure across providers and 1 indicates one provider capturing all of it, and the *equity ratio*, a provider’s

share of exposure divided by their share of the catalog, which functions as our rich-get-richer check. *Catalog share* and *provider exposure counts* serve as supporting detail beneath these two headline measures. These are measured as observed outcomes of each model and intervention, not optimized; no equity term enters any model objective, and we did not select on equity criteria.

Results

Mode A: User-to-Item

Baselines: ALS scores exactly 0 on both $\text{NDCG}@100$ and $\text{HitRate}@100$ at $k=0$, since it has no way to rank an item with no interaction history. CBHCF enables ranking immediately, reaching an $\text{NDCG}@100$ of 0.0419 and a $\text{HitRate}@100$ of 0.1490 at zero interactions.

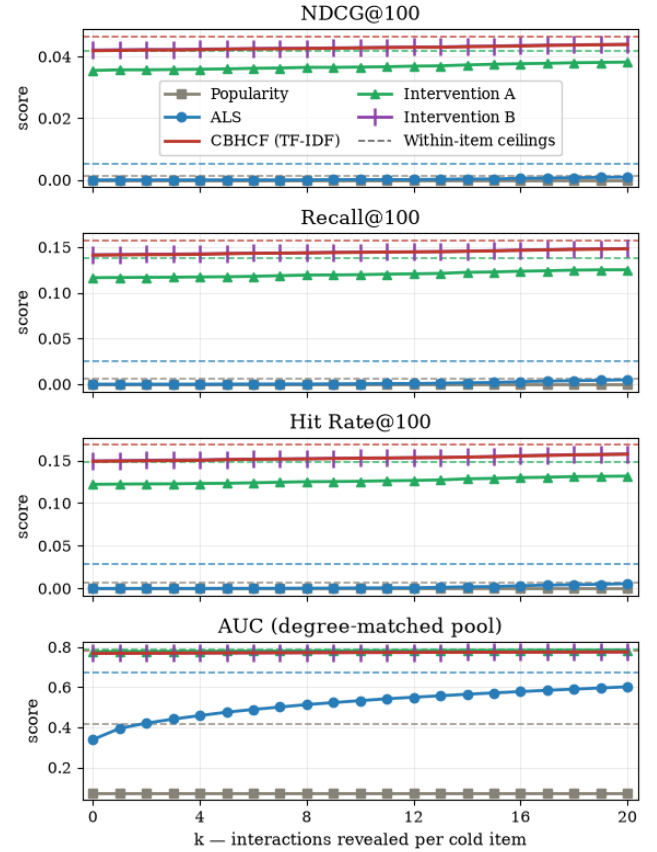


Figure 2. Mode A Warm-up Curves

Intervention A: Intervention A, which replaces CBHCF’s TF-IDF representation with semantic embeddings (i.e., representing text based more on overall meaning rather than exact word overlap), does not improve top-100 retrieval results. Its $\text{NDCG}@100$ increases from 0.0355 at $k=0$ to 0.0382 at $k=20$ (+7.6%), but remains below CBHCF across the warm-up curve.

Intervention B: Intervention B folds in seven additional synthetic interactions at each reveal level. Intervention B’s $\text{NDCG}@100$ rises from 0.0421 to 0.0440 across the warm-up curve, compared with 0.0419 to 0.0439 for CBHCF. We control the LLM-selected interactions against randomly-selected interactions from the same candidate pool. The LLM’s ordering beats random by a small but statistically significant margin, suggesting it adds signal, but

the effect size is negligible. Selecting the most active users beats both.

Overall, neither intervention changes the shape of the Mode A warm-up curve. Each gains roughly the same absolute amount over 20 reveal levels as CBHCF, only shifting the level at which the curve sits.

Mode B: Item-to-User

Mode B ranks users for an item and asks whether the item's ranking surfaces the users who actually want it.

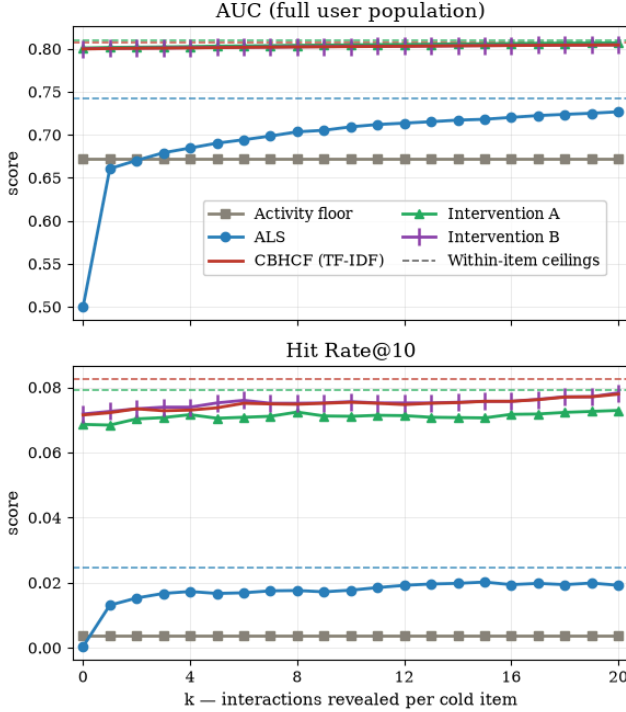


Figure 3. Mode B Warm-up Curves

Baselines: The Activity baseline (recommending the most active users) reaches AUC = 0.6724 while retrieving almost nothing at K=10 (HitRate@10 = 0.0037), reflecting how Mode B AUC is not degree-matched like Mode A AUC. Heavy users are disproportionately likely to be in the test set, so 0.6724, not 0.500, is the effective baseline for reading other models' Mode B AUC.

At k=0, pure CF scores an AUC of 0.5000. This is because without interaction history, every user gets the same score and no ordering is possible. As interactions accumulate, ALS improves, reaching an AUC of 0.7268 after 20 interactions and 0.7428 at its within-item warm ceiling. Even then, that is only 0.0704 above the Activity floor. In contrast, CBHCF achieves an AUC of 0.7998 at zero interactions, 0.1274 above the floor, indicating that item content alone provides a strong ranking signal.

Intervention A: Intervention A produces slightly higher overall ranking performance (AUC) but lower top-10 retrieval performance. At zero interactions, Intervention A achieves an AUC of 0.8009 compared with 0.7998 for CBHCF; however, its HitRate@10 is 0.0686 compared with 0.0715 for CBHCF. After 20 interactions, Intervention A reaches a HitRate@10 of 0.0729, while CBHCF reaches

0.0779, a gap of roughly ten times the seed standard deviation of 0.0005 for Hit Rate (see Appendix A#Reproducibility for error band information). As such, semantic representation shows slight improvement in the overall ordering of users but degrades the model's ability to place the relevant users within the top 10.

Intervention B: Intervention B provides no measurable change on Mode B relative to the CBHCF baseline. At k=20, the reasoning arm scores an AUC of 0.8047 compared to a baseline of 0.8045 for CBHCF, and a HitRate@10 of 0.0783 compared to 0.0779. The HitRate@10 gap is 0.0004, smaller than the seed standard deviation of 0.0005. Furthermore, the random control scores 0.8046 and 0.0781, so the LLM ordering is not distinguishable from random selection. Even at k=0, adding seven synthetic interactions only moves AUC from 0.7998 to 0.8001, indicating that the synthetic collaborative signal is negligible compared to the content signal. As such, we have no evidence that Intervention B helps in Mode B. However, we did not analyze Mode B with the same paired per-seed test we used for Intervention B in Mode A, so there's a possibility that a real but small effect exists.

Provider Equity

The catalog itself is not evenly distributed across providers (Gini=0.554), so that value (not zero) is the reference value. Every model concentrates exposure more.

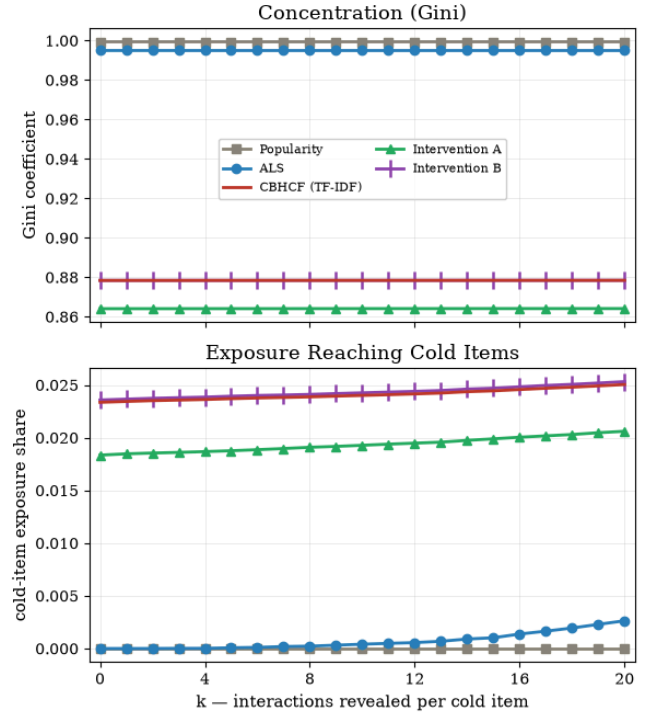


Figure 4. Provider Exposure Equity Curves

Popularity is the most concentrated (Gini = 0.9994), followed by ALS (0.9950). The content-based models are flatter, with CBHCF at 0.8784 and Intervention A producing the lowest concentration at 0.8641. Intervention B matches CBHCF at 0.8784, indicating that folding in synthetic interactions does not change which providers received

exposure. Gini is effectively constant across the warm-up curve.

It is important to note that lower concentration does not necessarily correspond to greater total exposure for cold items. Intervention A distributes exposure more evenly than CBHCF (Gini of 0.8641 vs 0.8784) but provides cold items a smaller share of exposure (0.0206 vs 0.0250). Intervention B matches CBHCF on both. However, all three content-based approaches improve both metrics compared to the interaction-only and popularity baselines.

Discussion, Broader Impacts, Ethics, and Limitations

Our work could impact recommenders, providers, and consumers. For recommenders, the clearest result is that leveraging content representations enables an immediate but partial answer to the cold-start problem. For providers, improving the item metadata could help their cold items receive better recommendations immediately. In our study, adding TF-IDF content representations provided an immediate lift of 0.1490 in HitRate@100 and 0.0419 in NDCG@100 at zero interactions, against a pure CF baseline of exactly zero on both. CBHCF maintains this advantage and reaches 8.8x the NDCG@100 once warm. Providers can use this as further motivation to create quality metadata for cold items. Consumers can use this to gain a better understanding of how systems they use recommend items.

The study evaluated two interventions intended to improve how well a recommender system handles cold items. Intervention A replaced TF-IDF content features with dense semantic embeddings, and Intervention B used synthetic interactions as data augmentation. We evaluated recovery across a warm-up curve for both user-to-item and item-to-user recommendations. The results suggested that performance depended less on how quickly interactions accumulated and more on the quality of item representations. Consequently, shortening the warm-up period was not the most important finding; the stronger result was how well the system performed from the very beginning, at zero interactions.

Both language-model interventions added little beyond what item content has already provided, though for different reasons. Intervention B produced the highest NDCG in our comparison, 0.04206 compared with 0.04192 for CBHCF, a difference of only 0.33%. Randomly selecting seven users from the same content-filtered shortlist produced about two-thirds the effect size that Intervention B achieved. This suggests that the LLM's ranking contributed only about one-third of the improvement, or roughly 0.11% of the baseline. Whether the remaining gap reflects a genuine ceiling on synthetic interactions or simply the small number we synthesized, and selection method, we cannot say from this experiment. Intervention A changes representation rather than data volume, and the semantic embedding model we selected performed worse than the TF-IDF baseline on the main ranking measures. This does not establish that semantic embeddings are worse in general, only that this particular embedding choice underperformed here. MARec

[1] found embeddings improved HitRate@100 by ~23% over TF-IDF on MovieLens 10M.

The results also raise the question about which items benefit, and the pattern here is that models with stronger recommendation performance also gave more exposure to cold and less-popular items even though neither model was designed with fairness in mind. Under ALS, cold items captured only about 0.264% of exposure versus 2.504% under CBHCF and 2.532% under Intervention B. One plausible explanation is that content gives a new item useful information to compete with immediately. A book with only 27 reviews can still have a title, author, category, and description, just like a book with 1,759 reviews. However, greater use of content does not eliminate bias; it may change where the advantage comes from. Titles, item features, and feature descriptions, , and other forms of metadata are not equally detailed or well produced for every item. Furthermore, a large publisher may provide more polished descriptions and extensive marketing material than an independent author. If recommendation quality depends heavily on metadata, items with richer or better-written content may still have an advantage.

Ethics and Limitations

Several ethical considerations and limitations should be acknowledged in conjunction with our report.

Qwen2.5 may have trained on books and content from the Amazon Reviews dataset. We acknowledge that inferred preferences cannot be fully separated from the model's prior knowledge of the titles.

To measure a warm-up curve over 21 reveal levels ($k=0$ to 20) with a test set of 5, we restricted cold-item eligibility to items with 25+ interactions. Therefore, our cold population is not representative of genuinely obscure items.

Author-name tokenization was done without access to an authoritative list, so "Stephen King" may appear differently from "King, Stephen", and "Harper, Lee" may tokenize the same as "Lee, Harper".

Intervention A only changes how content is represented, and CBHCF's content term is static with respect to how many interactions have been revealed for an item. Therefore, Intervention A can shift the level of cold-start performance, but it cannot significantly change the shape of the warm-up curve.

Intervention B does not follow ColdLLM as published [13]: we use an off-the-shelf 7B model without fine-tuning, and a simplified filtering stage, calculating similarity between a user's content history and the item in question's content vector, rather than a two-tower model. Furthermore, the effect is real but practically negligible, so our claim that the LLM adds value is deliberately narrow. Since Intervention B injected 19,089 synthetic interactions against roughly 2.67 million real ones, our result may be bounded by inadequate volume rather than a general finding. Finally, because the content filter selects candidates with no reference to the train/test split, a synthetic pair can coincide with a real test interaction.

There is a risk our synthetic data misrepresents user preferences. Intervention B should be viewed only as a modeling technique since the synthetic preferences were not actually expressed by users.

Equity measurement is limited by the lack of impression logs. Since Amazon Reviews does not have exposure data, the position discount used to weight exposure by rank is an assumption that may not align with reality. Furthermore, equity was measured on eval users only, not the full population. Finally, using a deterministic top-K inflates measured inequality, but we keep it because a stochastic ranking policy would change our accuracy metrics.

Across models, exposure repeatedly favored the same authors, and we acknowledge that this form of recommendation bias may persist despite our interventions.

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Statement of Work

All team members collaborated throughout the project by reviewing key decisions, sharing feedback, troubleshooting challenges, and supporting the integration of each component into a cohesive final system. Individual contributions are summarized below:

Freya Van de Motter
Led project scoping, definition of the cold-start task, architecture design, collaborative filtering development, cold-user and cold-item logic, and construction of the warm-up evaluation framework.
Jessica Lan
Led repository configuration, experiment-tracking research and implementation, reproducibility infrastructure, and development of the content-aware hybrid component.
Kirsten Emerson
Led the evaluation framework, definitions of warm, sparse, cold-user, and cold-item groups, ranking metrics, failure analysis, exposure-equity analysis, benchmark tables, and results visualization.
Seema Guruvadoo
Led data ingestion, filtering, cleaning, metadata preparation, interaction standardization, exploratory analysis, and documentation of the data pipeline.

Use of Generative AI

Artificial intelligence was used in two distinct ways. First, pretrained language models were part of the experimental recommendation system. Intervention A used a pretrained text-embedding model (snowflake-arctic-embed-l-v2.0) to generate embeddings, and Intervention B used a generative language model (Qwen-2.5-7B-Instruct-AWQ) to create structured synthetic preference signals for items without observed interaction histories.

Second, generative AI tools were used during project development for code assistance, debugging, and documentation architecture brainstorming. AI-generated code was not accepted without review. Team members inspected the outputs, tested code against the project data, corrected incompatible assumptions, and take responsibility for all methodological decisions and claims.

Appendix A: System Description

The Cold Start Problem and How We Measure

An item is **cold** when it has no interaction history. Pure collaborative filtering has no signal to work with when items are cold, because it infers preference solely from interaction history. To combat this, **content-based hybrid collaborative filtering** algorithms have been developed that leverage the information that an item ships with to improve filtering for both cold and warm items. The **warm-up curve** shows how quickly a system recovers as interactions arrive.

Our research question is whether **language models** can be leveraged to speed up the warm-up curve, and how doing so affects **provider-side equity**. We evaluate along a **reveal level** k . At $k=0$, the model sees no interactions for these items. At $k=20$, the model sees all of the revealable pool. Therefore, our metrics are curves over $k=0\ldots20$, not a single number.

In our model, revealing an interaction does not retrain the model. The model is fit once, and each cold item is **folded in**, meaning its own representation is recomputed from the k revealed interactions while every user’s learned preferences stay frozen. Since only the item representation moves, the curve isolates the effect of the accumulating interaction history.

We include two references with every curve. The **floor** is a naïve model that ignores personalization. The **ceiling** is the same cold items folded in with their complete pre-test history and scored on the same reserved test set following the fold-in evaluation protocol used in [2, 6]. For an item with exactly 25 interactions, the ceiling is the same as the $k=20$ endpoint. Items with more than 25 interactions fold in everything before the reserved test set, so their ceiling sits above that endpoint.

Using the within-item ceiling allows us to do three things. First, it tells us what share of achievable performance we capture at each step of the warm-up curve. Next, it shows us what success would look like, since this performance gap is what the project is trying to close. And finally, it rules out the alternate explanation that the cold items are simply harder to score, since both sides of the comparison are the same items.

The Data

The Amazon Reviews dataset, including both user reviews and item metadata, was collected by the UC San Diego McAuley Lab. The datasets are publicly available and can be accessed from the official Amazon Reviews website and the Hugging Face Datasets hub. We also include links in the README of our accompanying GitHub repository.

We use the Amazon Reviews 2023 Books dataset, restricted to its 5-core subset and further restricted to users who also reviewed in the Amazon Reviews 2023 Movies & TV dataset [10]. That cross-category restriction kept a cross-domain cold start option open, but we did not get to that point and no Movies & TV items, interactions, or metadata enter any of our models. What remains are

roughly 5.1 million interactions over 487,790 items and 384,339 users. Only ratings of 4 stars or more counted as a positive interaction. After that filter and the warm/cold split, models train on roughly 2.67 million interactions and rank over a candidate pool of 246,687 items.

Only items with 25+ interactions are eligible to be treated as cold, and have a median of 38 interactions. We split this subset of 27,268 items 80/10/10 into a warm set, a cold validation set, and a cold test set, with 2,727 cold items in the validation and test sets. All items with 5-24 interactions are then added to the warm set, and items below 5 are dropped. For the validation and test sets, the first 20 interactions form the revealable pool with which we measure the warm-up curve and the final five interactions form the test set.

All warm items are trained on, except that each user’s last warm interaction is used as a warm test case and their second-to-last as a warm validation case. We use these during hyperparameter tuning to ensure we do not select a configuration for cold-start performance that greatly degrades ordinary warm recommendations. Hyperparameter tuning and configuration choice are made on the cold validation population, and the cold test population is used for final results only. Users with fewer than three warm interactions have everything put in training, and items with fewer than five interactions are dropped before the split.

Two Modes

The system is evaluated from two directions, ranking along both axes of the same user-by-item matrix.

	Mode A	Mode B
Question	Did the cold item reach this user’s top-K?	Did the right users reach the item’s top-K?
Ranks	items, for a user	users, for an item
K	100	10
Floor	Popularity	Activity

Table A.1: Mode Descriptions

Each has separate floors and ceilings, and their metrics rank against different candidate sets, so **numbers from one mode cannot be compared to numbers from the other**.

Mode A has an asymmetry: the top-K metrics (NDCG, Recall, Precision, and Hit Rate) rank against the full candidate pool [14] (chosen over sampled negatives, which can lead to worse models scoring higher [15]) but AUC ranks against a degree-matched pool restricted to warm items with at least 25 training interactions. Without that restriction, the Popularity baseline earns a misleadingly high AUC simply by beating the long tail of warm items (whose median degree is 7 interactions), a popularity artifact of the kind described in [16, 17].

Mode B’s AUC ranks against the whole user population, with no degree matching. We downsample each cold item’s held-out users to five, so Precision and Recall are not dominated by a handful of high-degree items. This caps Precision@10 at 0.5 by construction.

The Reference Models

We create three baseline models per mode, each one adding capability above the last.

Popularity (Mode A) and **Activity** (Mode B) are the naïve floors. Popularity ranks items by global interaction count (how *popular* they are). Activity ranks users by global interaction count (how *active* they are). Neither is personalized. They establish a reference point for the metrics. Without them, an NDCG of 0.04 would be uninterpretable.

ALS (Alternating Least Squares) is the pure collaborative filtering model [18], which demonstrates what the interaction signal can achieve by itself. ALS cannot rank items with zero interactions. Without interactions, there’s nothing to represent the item with, so cold items are scored identically and none of them reach the top-K of any user’s list ALS was chosen over another popular CF algorithm, Bayesian Personalized Ranking (BPR) [19], because ALS supports fold-in (which we use in calculating the warm-up curve) while BPR does not.

CBHCF (content-based hybrid collaborative filtering) is our **true baseline**. Both interventions are measured against this. CBHCF’s score for a user and an item is the sum of two parts: the collaborative score from ALS, and a content score that measures how similar a given item is to other items the user already engaged with. A parameter, lambda, sets the weight of the content score. Content similarity is calculated by comparing TF-IDF vectors of the book’s title, author, genre, item features, and item description text, consistent with prior work on Amazon Reviews data [1, 7]. Each field is vectorized separately and has a per-field weight, so long fields cannot swamp short ones, grounded in [20]. This content score allows cold items to be ranked based on what they are before interaction history appears.

CBHCF reuses the same fitted ALS model, which is what makes the ALS-to-CBHCF gap attributable to the content score alone. It also builds each user’s content profile from their warm training history and never updates it. Therefore, the user factor from ALS and the user profile from CBHCF are both frozen as interactions are folded in. Since ALS contributes exactly zero at $k=0$, a cold item is ranked on content alone at the start of the CBHCF curve.

The Interventions

Each intervention is a single-variable substitution on top of CBHCF, designed so that any difference in the curve is attributable to that change alone.

Intervention A: Dense Semantic Embeddings

Intervention A uses **dense semantic embeddings in place of lexical** (TF-IDF) **vectors** for the prose fields (title, item features, item description). A sentence embedding model, **snowflake-arctic-embed-l-v2.0**, did the encoding. It was selected on the cold validation population from a slate of six candidates. Entity-like fields (author and genre) were held lexical throughout, on the grounds that exact matching is the correct semantics for a controlled vocabulary. We did not test that assumption directly, but followed learnings in MARec [1]. Everything else was held fixed: the same ALS

fits, the same frozen user preferences, the same lambda mechanism, and the same evaluation path. Both CBHCF and Intervention A had their field weights and lambda searched on an identical grid with four values for the combined title-and-publisher-blurb block, four for the editorial review block, and three for lambda, with author and genre held fixed in both. Therefore, neither entered the comparison with a tuning advantage.

Intervention B: Synthetic Interactions

Intervention B follows the ColdLLM method [13] in leveraging a two-stage funnel to generate synthetic but plausible interactions for cold items. First, a content-similarity filter identified the fifty users whose reading history most closely resembles the cold item's text. An LLM (**Qwen2.5-7B-Instruct-AWQ**) then scored each surviving (user, item) pair, and the top seven per item became synthetic interactions. We chose seven as that was the median warm item's training degree. This results in 19,089 synthetic interactions.

We compared two prompting strategies: asking the LLM for a direct verdict, and asking it for a one-sentence reason for its verdict first. The direct prompt answered yes for 97.1% of the pre-filtered candidates, and the reasoning prompt answered yes to 88.3%. The ordering of "yes" candidates is determined by reading the LLM's log-probabilities at the yes/no answer token and then scoring the pair by its log-odds there (log probability of yes minus log probability of no).

Those interactions were folded-in at every reveal level including 'k=0'. The performance was matched against controls that injected the same number of synthetic interactions. The first control selected at random, which allowed us to isolate the value contributed by the LLM by virtue of its ordering (within the filtered candidate pool) versus how much the score improved just by virtue of the content-similarity filtering in the first stage. The other selected by popularity, which tested whether the LLM's ordering was a proxy for user activity.

Because the content filter selects candidates with no reference to the train/test split, a synthetic pair can coincide with a real interaction in the test set. About ~1% of the injected pairs were in the test set, equalling ~1.4% of the test set.

Note - ColdLLM [13] adds synthetic interactions to training and refits. Refitting perturbs user factors as well as item factors, which goes against our design principle that user preferences are held constant across every condition. Therefore, we allowed synthetic interactions to enter only during fold-in. We tested both approaches and they agreed to five decimals. Since production systems are not retrained every time a new item is added, fold-in is also closer to how such a system would be deployed in practice [13].

We found that the reasoning prompt reliably produced roughly twice the effect of the direct prompt across ALS random seeds. However, both effect sizes are very small in absolute terms. The LLM's ordering is real signal, but does not meaningfully improve cold-start performance. Furthermore, since we tracked the user-degree of the users it

recommended, we were able to determine that the users recommended via LLM ordering have lower average degree than those selected by random, even though it has roughly twice the effect size of random. Therefore, we can also determine that the LLM is not a proxy for user activity.

Intervention B's two prompting strategies and both controls were run in Mode A, but only the reasoning variant and the random control were carried to the steel thread and reported in both modes.

Provider Equity

Accuracy metrics ask if cold items become retrievable. Provider equity asks whose items are being recommended, and whether exposure is being concentrated on providers who are already well represented.

A provider is the entity credited with creating an item (for Books, the author) and is the unit across which we measure exposure concentration, since it is providers rather than individual titles that gain or lose from a recommender system's behavior. We used the same function that built the author field in the content-based models to determine provider identity. Our catalog of 246,687 items has 86,791 providers.

All metrics are computed over the same population: the 12,382 users who have at least one cold item in their held-out test set.

Exposure is our measure of visibility. An item's exposure is the sum of its appearances across all users' top-100 recommendations, each weighted by position using a log-discount. A provider's exposure is the sum of its items'. The log-discount is the same discount NDCG uses, so the accuracy and equity curves weight positions identically.

We report three measures. *Gini* is computed over the full provider universe, including zero-exposure providers. The *share of exposure going to cold items* is the exposure of all cold items divided by the exposure of all items. The *cold equity ratio* divides that share by all cold items' share of the catalog (per-provider, weighted by provider exposure), which is the measure most directly related to our research question. A ratio of 1 means cold items receive exposure in proportion to their presence. Above 1 means overexposed and below 1 means underexposed.

The Gini score over the same 86,791 providers and 246,687 candidates is 0.554, whereas exposure Gini runs from 0.86 for Intervention A to 0.995 for ALS, and 0.9994 for the Popularity baseline. Every model concentrates exposure well beyond the concentration already present in the catalog. We therefore compare models against each other rather than reading the absolute numbers. Every model uses the same ten ALS fits and the same frozen user preferences, so the difference between them comes from item representation alone.

Reproducibility

Every ALS-based approach reuses the same 10 fitted ALS models (which get initialized with different random seeds and are otherwise identical), which allows us to

mitigate the risk of differences in model initialization skewing our results. The band on every curve is the spread across those ten seeds, i.e. sensitivity to model initialization. Popularity and Activity are deterministic and have no error band.

Configuration selected during the project is written to `outputs/hyperparams.json`, and stamped with a fingerprint of the data split so that rerunning updated code does not silently reuse caches created with deprecated splits.